Bharat Ratna Indira Gandhi College of Engineering, Solapur

Master of Computer Application

SUBJECT:

FULL STACK DEVELOPMENT (MCAC301)

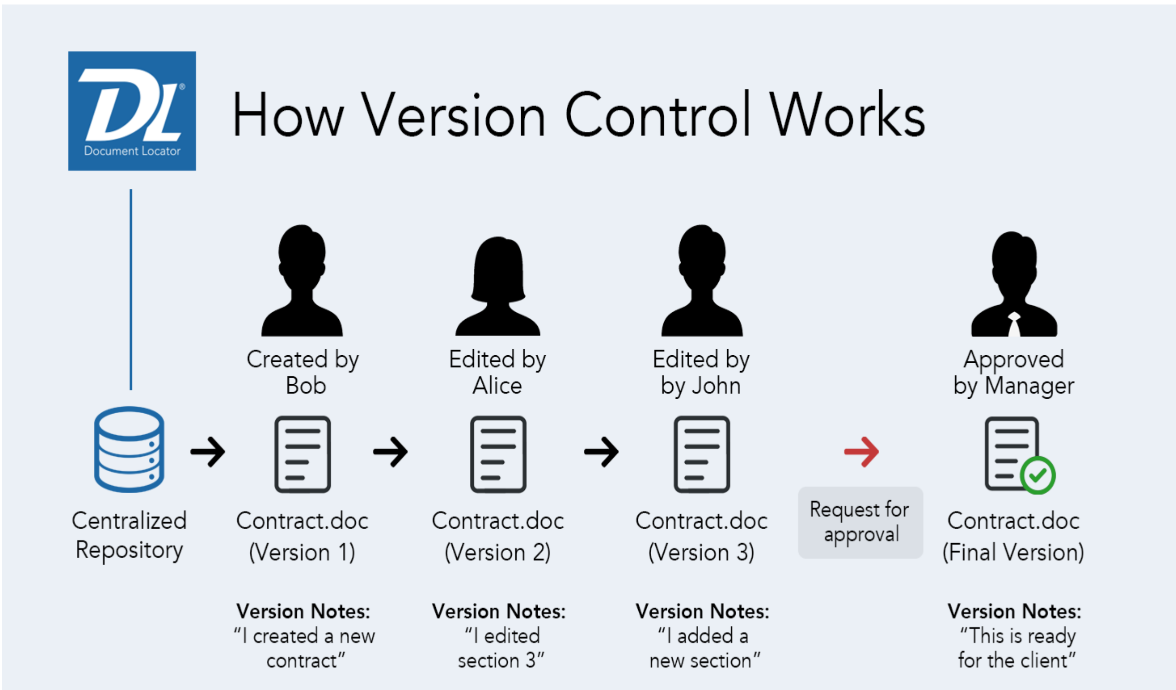
SEMESTER-III

UNIT 3: INTRODUCTION TO VERSION CONTROL

Introduction to Version Control

What is Version Control?

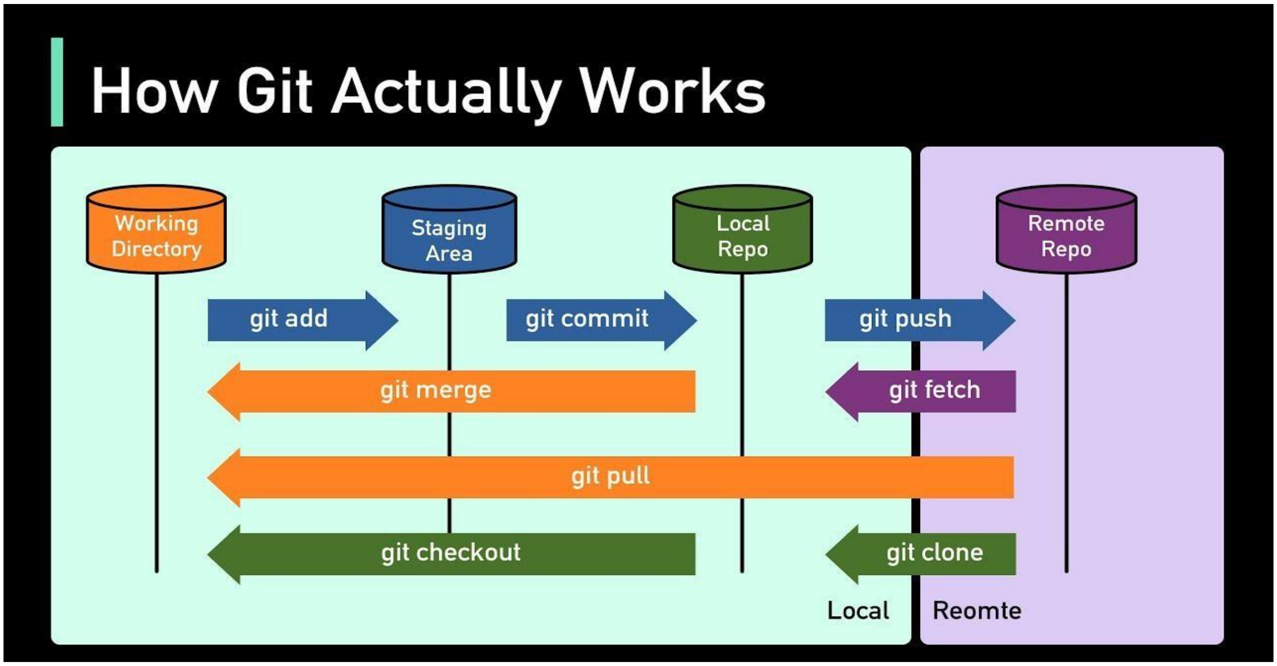
- **Definition**: A system that records changes to files or sets of files over time, allowing you to revert to previous versions, track changes, and collaborate with others.
- **Purpose**: To manage changes, track history, and facilitate collaboration among multiple developers.



Unit 1: Git

What is Git?

- **Definition**: A distributed version control system that tracks changes in source code during software development.
- **Developed by**: Linus Torvalds in 2005.
- **Key Features**:
  - Distributed architecture
  - Branching and merging
  - Strong support for non-linear development



Basic Git Commands

- **Initialize Repository**: Create a new Git repository.

```
git init
```

- **Clone Repository**: Create a local copy of a remote repository.

```
git clone <repository-url>
```

- **Check Status**: Show the working tree status.

```
git status
```

- **Add Files**: Stage changes for the next commit.

```
git add <file-name>

git add .
```

- **Commit Changes**: Record changes to the repository.

```
git commit -m "Commit message"
```

- **View Commit History**: Display the commit history.

```
git log
```

- **Create Branch**: Create a new branch.

```
git branch <branch-name>
```

- **Switch Branch**: Change to a different branch.

```
git checkout <branch-name>
```

- **Merge Branch**: Merge changes from one branch into another.

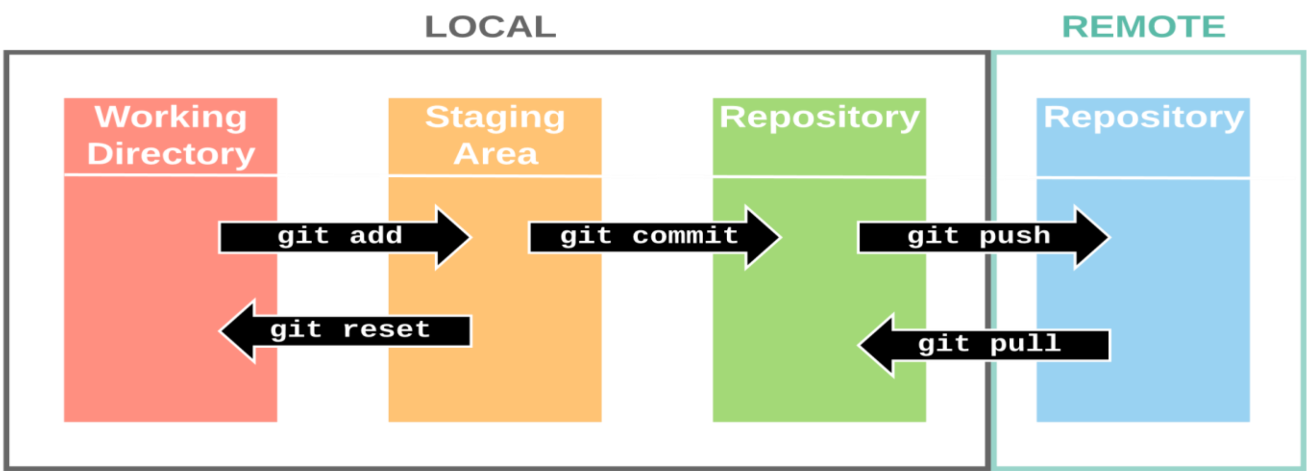
```
git merge <branch-name>
```

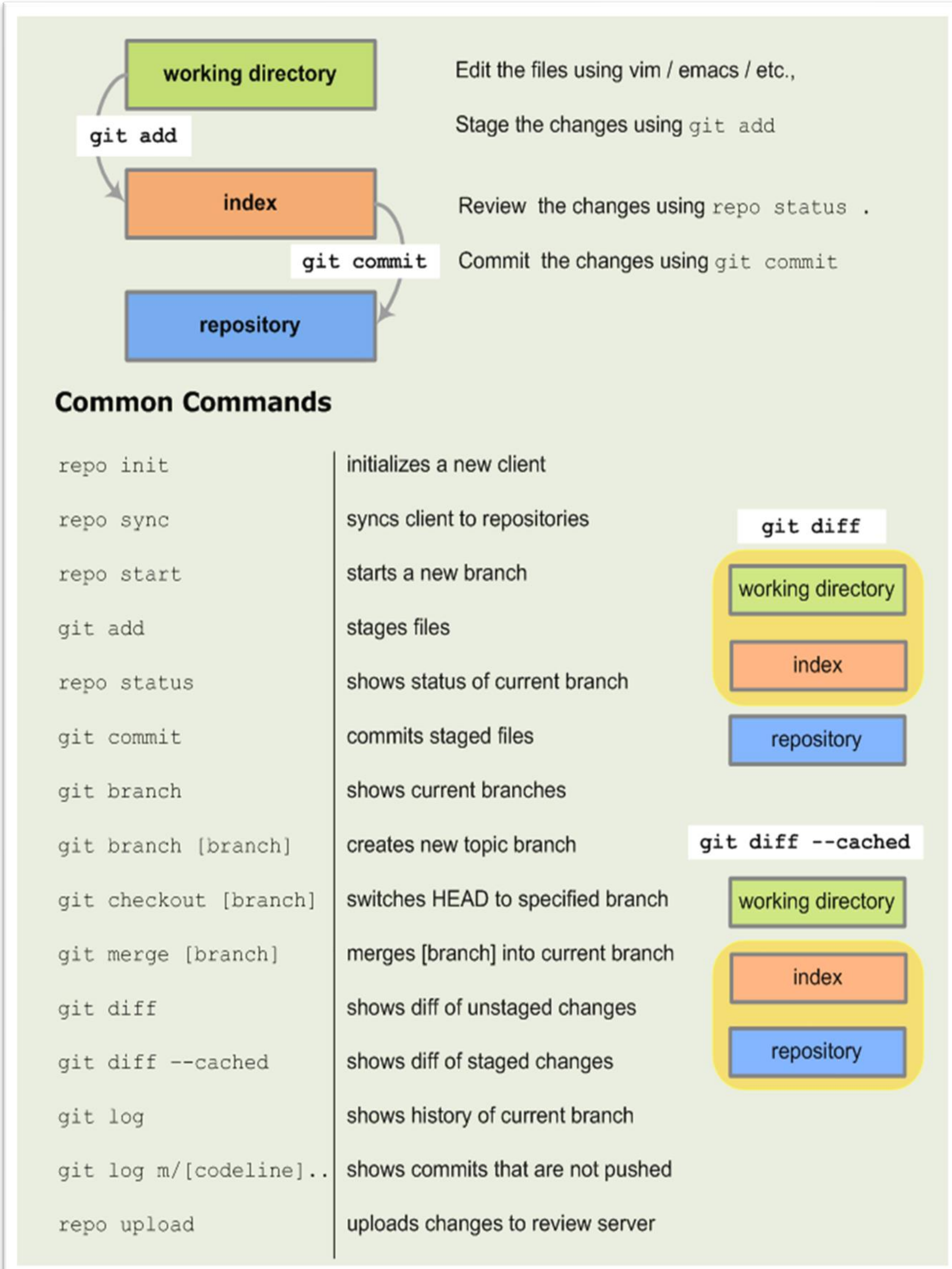
- **Push Changes**: Upload local changes to a remote repository.

```
git push origin <branch-name>
```

- **Pull Changes**: Fetch and merge changes from the remote repository.

```
git pull
```




working directory
git add
index
git commit
repository
Edit the files using vim / emacs / etc.,
Stage the changes using git add
Review the changes using repo status .
Commit the changes using git commit
Common Commands
repo init
initializes a new client
repo sync
syncs client to repositories
repo start
starts a new branch
git add
stages files
repo status
shows status of current branch
git commit
commits staged files
git branch
shows current branches
git branch [branch]
creates new topic branch
git checkout [branch]
switches HEAD to specified branch
git merge [branch]
merges [branch] into current branch
git diff
shows diff of unstaged changes
git diff --cached
shows diff of staged changes
git log
shows history of current branch
git log m/[codeline]..
shows commits that are not pushed
repo upload
uploads changes to review server
git diff
working directory
index
repository
git diff --cached
working directory
index
repository

Parameters	GitLab	GitHub
Developed by	GitLab was developed by Dmitriy Zaporozhets and Valery Sizov.	GitHub was developed by Chris Wanstrath, Tom Preston-Werner, P. J. Hyett, and Scott Chacon.
Open source	GitLab is open-source for community edition.	GitHub is not open source.
Public Repository	It allows users to make public repository.	It allows users to have unlimited free repository.
Private Repository	GitLab also provides free private repository.	GitHub allows users to have free private repository but with maximum of three collaborators.
Navigation	GitLab provides the feature of navigation into the repository.	GitHub allows user to navigate usability.
Project Analysis	GitLab provides user to see project development charts.	GitHub doesn't have this feature yet but they can check the commit history.

Unit 2: GitHub and GitLab

GitHub

- **What is GitHub?**
  A web-based platform that uses Git for version control and provides additional features like issue tracking, code review, and collaboration tools.
- **Key Features**:
  - Repository hosting
  - Pull requests and code reviews
  - Issues project boards

GitLab

- **What is GitLab?** A web-based DevOps platform that provides Git repository management, CI/CD, and other collaboration features.

- **Key Features**:
  - Repository hosting
  - Merge requests and code reviews
  - Project Tracking Management
  - Built-in CI/CD pipelines

Key Differences Between GitHub and GitLab

- **CI/CD Integration**: GitLab has built-in CI/CD capabilities, while GitHub relies on GitHub Actions or third-party services.

- **Pricing and Features**: Both platforms offer free and paid plans, with different feature sets. GitLab's free tier includes more built-in features, while GitHub's free tier offers integration with many third-party tools.

Unit 3: Code Reviews and Collaboration Workflows

Code Reviews

- **Purpose**: To ensure code quality, find bugs, and improve code readability and maintainability.

- **Best Practices**:

- Review code for functionality, performance, and security.
- Provide constructive feedback and ask for clarification when needed.
- Follow coding standards and guidelines.
- Use tools like GitHub Pull Requests, GitLab Merge Requests, or code review tools like Crucible.

Collaboration Workflows

- **Feature Branch Workflow**: Developers create a new branch for each new feature or bug fix, which is merged into the main branch after review.

Steps:

1. Create a new branch from the main branch.
2. Develop the feature or fix.
3. Push the branch to the remote repository.
4. Open a pull request or merge request for review.
5. Review and merge the branch into the main branch.

- **Git Flow**: A branching model with dedicated branches for features, releases, and hotfixes.
  Branches:
  - `master` (production-ready code)
  - `develop` (integration branch for features)
  - `feature/*` (feature branches)
  - `release/*` (release branches)
  - `hotfix/*` (hotfix branches)

	HELIX TEAMHUB	GITLAB	GITHUB	BITBUCKET
Side-by-side view	✓	✗	✗	✗
Quick Action Buttons	✗	✓	✓	✓
Supports adding images	✓	✓	✓	✓
Supports adding other type of attachments	✓	✓	✗	✗
Search functionality for wiki pages	✓	✗	✗	✗
Support for multiple markup languages	✗	✓	✓	✓
Possibility to view the history on code level	✓	✗	✗	✓

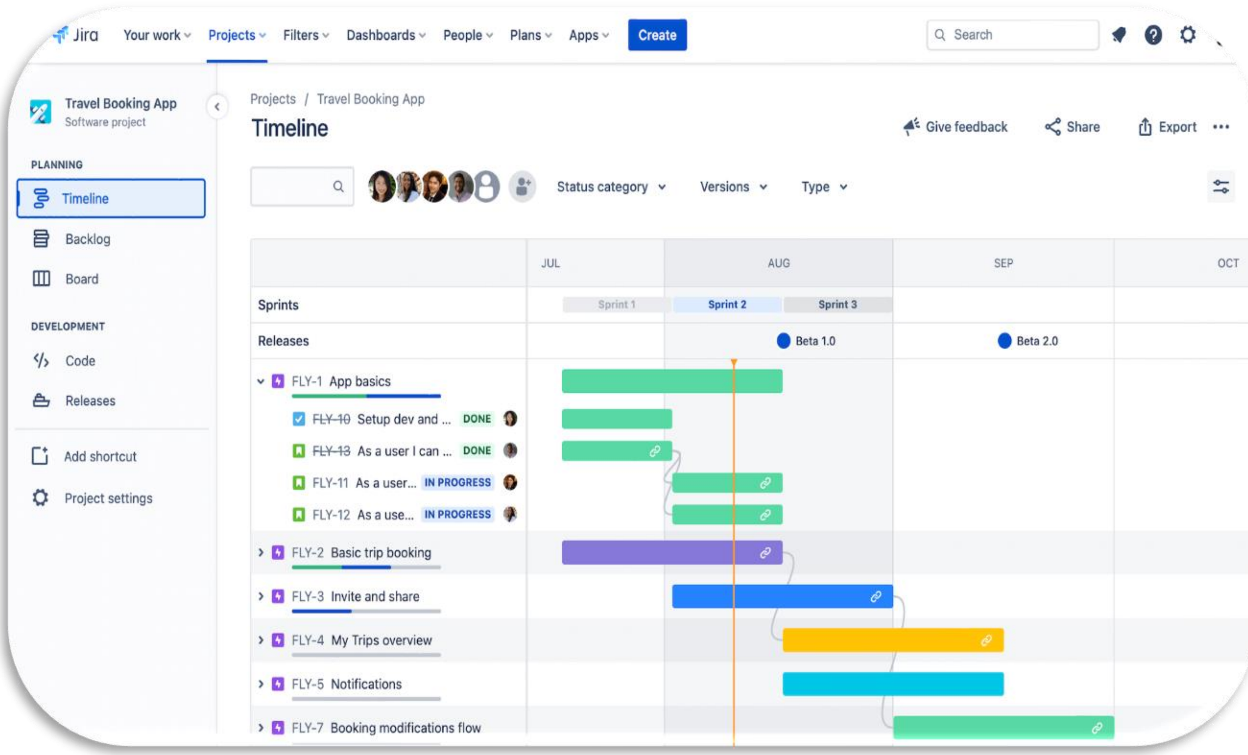
Unit 4: Issue Tracking and Project Management

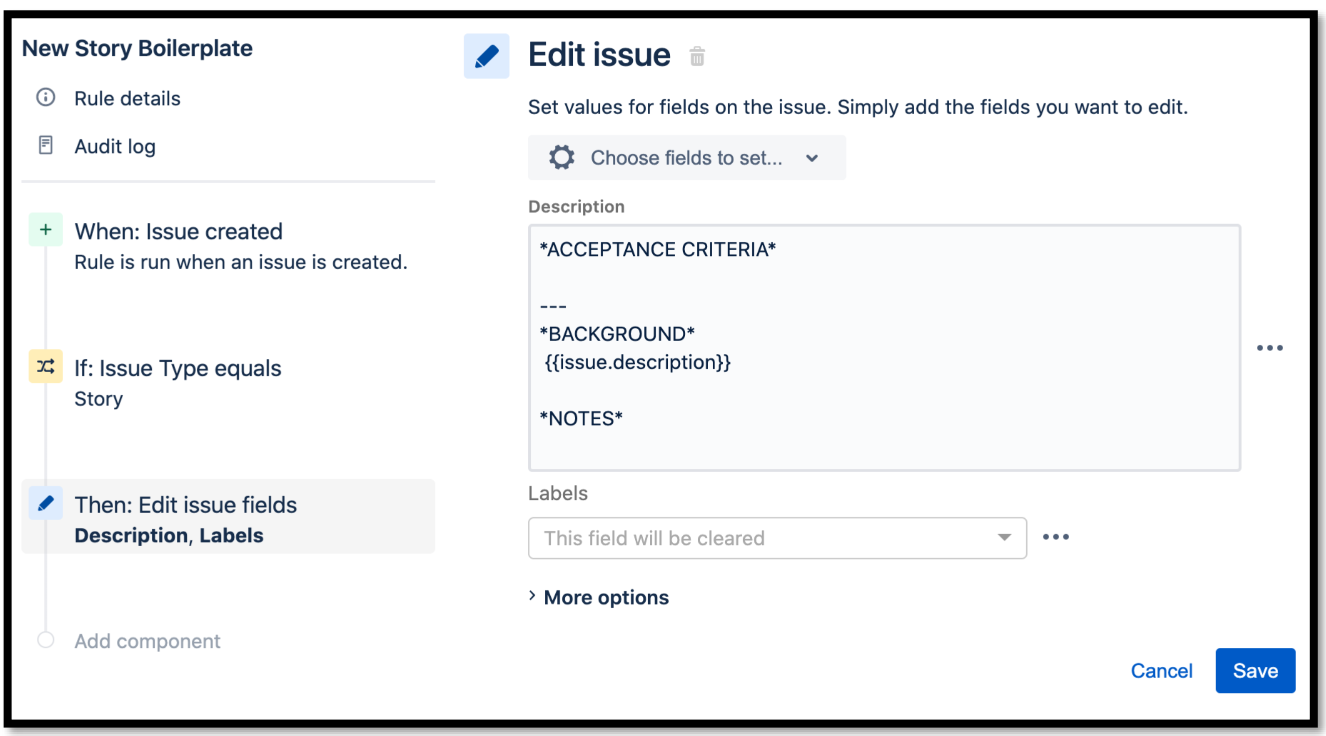
Issue Tracking

- **Purpose**: To track bugs, feature requests, and other tasks related to software development.
- **Tools**:
  - **GitHub Issues**: Integrated with GitHub repositories for tracking and managing issues.
  - **GitLab Issues**: Integrated with GitLab repositories, offering advanced tracking and reporting features.
  - **Jira**: A comprehensive issue and project tracking tool that integrates with many development tools.
  - **Trello**: A flexible, board-based project management tool that uses cards and lists to track tasks.

Jira

- **What is Jira?** A project management tool developed by Atlassian for tracking issues and managing agile projects.
- **Key Features**:
  - Issue and project tracking
  - Agile boards (Scrum, Kanban)
  - Custom workflows and dashboards
  - Integration with various tools (e.g., Bitbucket, GitHub, GitLab)
- **Description:**
  - JIRA is a bug tracking tool that allows software developers to plan, track and work faster. JIRA is the main source of information for future software release.
  - Developers can plan new features to be added and bugs to be fixed in the next release.
  - Organize Documentation Tasks & Track Documentation Progress


Jira
Your work
Projects
Filters
Dashboards
People
Plans
Apps
Create
Search
Travel Booking App
Software project
PLANNING
Timeline
Backlog
Board
DEVELOPMENT
Code
Releases
Add shortcut
Project settings
Projects / Travel Booking App
Timeline
Give feedback
Share
Export
Status category
Versions
Type
JUL
AUG
SEP
OCT
Sprints
Sprint 1
Sprint 2
Sprint 3
Releases
Beta 1.0
Beta 2.0
FLY-1 App basics
FLY-10 Setup dev and ... DONE
FLY-13 As a user I can ... DONE
FLY-11 As a user... IN PROGRESS
FLY-12 As a use... IN PROGRESS
FLY-2 Basic trip booking
FLY-3 Invite and share
FLY-4 My Trips overview
FLY-5 Notifications
FLY-7 Booking modifications flow


New Story Boilerplate
Rule details
Audit log
When: Issue created
Rule is run when an issue is created.
If: Issue Type equals
Story
Then: Edit issue fields
Description, Labels
Add component
Edit issue
Set values for fields on the issue. Simply add the fields you want to edit.
Choose fields to set...
Description
ACCEPTANCE CRITERIA

BACKGROUND
{{issue.description}}
NOTES
Labels
This field will be cleared
More options
Cancel
Save

Jira vs Trello:

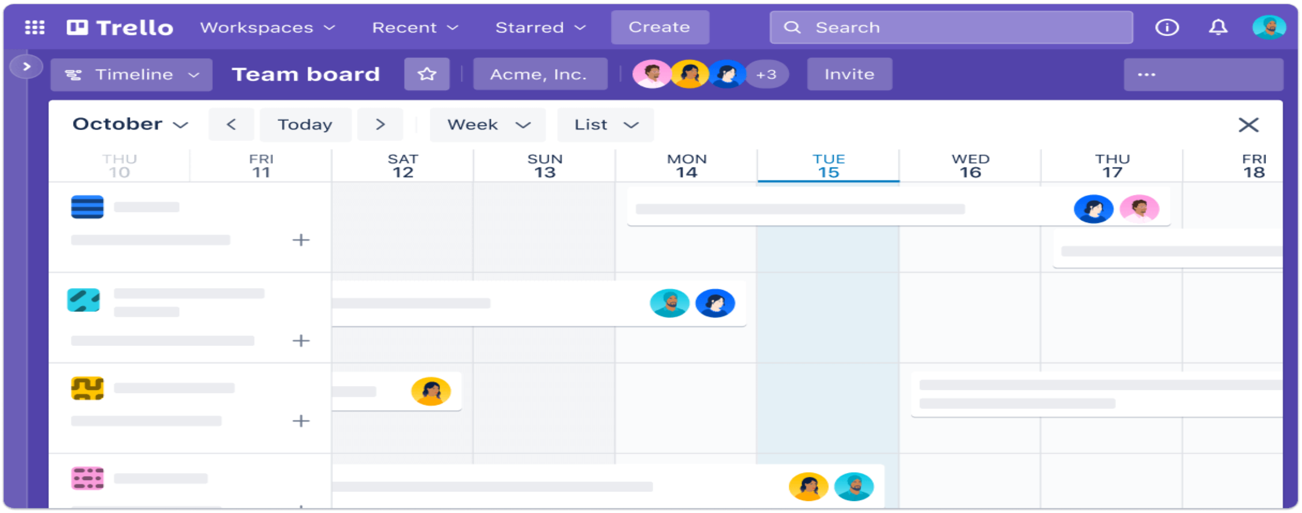
	Jira Software	Trello
TASK MANAGEMENT	✓	✓
TIME TRACKING	✓	✗
RESOURCE MANAGEMENT	✓	✓
REPORTING	✓	✗
DOCUMENT MANAGEMENT	✓	✓
ISSUE MANAGEMENT	✓	✗

Trello

- **What is Trello?**
  A visual project management tool that uses boards, lists, and cards to organize tasks.

Key Features:

- Drag-and-drop interface for managing tasks
- Customizable boards, lists, and cards
- Integration with various third-party tools and services
- Power-ups for additional features (e.g., calendar, automation)


Trello
Workspaces
Recent
Starred
Create
Search
Timeline
Team board
Acme, Inc.
+3
Invite
October
Today
Week
List
THU 10
FRI 11
SAT 12
SUN 13
MON 14
TUE 15
WED 16
THU 17
FRI 18